



CSE 2215 DSA - I

Assignment 1

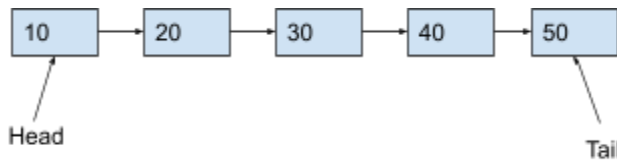
- Answer all the questions.
- The solutions must be handwritten.
- After the completion of the assignment , assign page numbers to every page.
- Scan every page, attach a cover page (with your name and id) and generate a pdf file
- Submit the assignment in the eLMS portal

SN	Problem
1	How does the <i>descending</i> order Merge Sort algorithm work on the following data? Draw the complete recursive tree. 33 42 12 8 28 102 78
2	Discuss the Drifting Queue Problem with the help of an illustrative example
3	Determine the time complexity of the following algorithm. sum=0; for(i=1; i<=n; i++){ for(j=1; j<=i; j++){ sum=sum+i+j; } } printf("%d", sum);
4	Answer the following questions for the doubly linked list as shown below, where p = last two digits of your student id + 5, q = p+3, r = p+q, s = r-1, t = r+s. <pre>graph LR head --> p p --> q q --> r r --> s s --> t t --> null null --> p temp --> r</pre>

- a) head -> next -> next-> value = ?
- b) last -> prev -> next->value = ?
- c) temp -> prev -> prev -> prev = ?
- d) temp -> next-> prev ->prev->value = ?
- e) last -> prev -> prev ->next-> value = ?

✓ 5

Consider the following single linked list.



Write the code for a function called **insert_before**. If you want to insert another Node before the Node 30 with the value 25 in the given linked list, the function can be called. The function will take two integers as parameters which in this example will be 30 and 25 respectively. After the function call **insert_before(30,25)** the condition of the linked list will be as follows:

